

Steven Spencer

Senior AI/ML engineer

✉ goldai.genius@gmail.com

in linkedin.com/in/steven-spencer-2603b7304

Profile

Experienced Senior AI/ML Engineer with over 12 years of expertise in designing, deploying, and maintaining AI and machine learning models across various domains. Proven ability to integrate AI/ML into real-time and batch applications, optimize algorithms, and enhance business processes. Strong background in Python, NLP, Generative AI, and machine learning frameworks like TensorFlow and PyTorch. Skilled in aligning AI solutions with regulatory compliance and operational excellence.

Professional Experience

2021/11 – 2024/12
San Antonio, Texas,
United States

Senior ML engineer

Polymer Physics Consulting, LL

- Designed, deployed, and maintained advanced machine learning models for industrial inspection systems, increasing defect detection by 40%.
- Built NLP models to analyze material behavior and perform text analysis, enhancing prediction accuracy by 30%.
- Integrated machine learning solutions into existing systems to optimize operational efficiency and reduce downtime by 25%.
- Engineered automated calibration solutions, cutting calibration time by 50% and ensuring compliance with strict standards.
- Conducted rigorous cost-benefit analyses using machine learning, reducing operational costs by 20%.

2018/01 – 2021/08
San Antonio, Texas,
United States

Machine Learning Engineer

MEDICAL & RADIATION PHYSICS, INC.

- Developed a strong foundation in electrical engineering through coursework in Electricity & Magnetism, Electromagnetics & Transmission Lines, and Power System Design, achieving a top 10% class ranking.
- Participated in cutting-edge research in computer vision, contributing to a project that improved image recognition accuracy by 15%.
- Designed, conducted, and analyzed optical systems, leading to a 20% enhancement in system performance and efficiency.
- Defined quality and performance metrics for optical design, resulting in a 25% reduction in defects and improved product reliability.
- Researched and prototyped optical setups, which led to a 30% increase in experimental throughput and data accuracy.
- Applied engineering principles to solve technical issues in radiation scanning systems, reducing false positive rates by 40%.
- Quantified costs and benefits for design solutions, effectively organizing and presenting results that led to a 15% cost savings in project implementation.

2015/10 – 2017/12
Sweetwater, Texas,
United States

Senior Machine Learning Engineer

Sauer Group, LLC

- Developed a machine learning-powered retail personalization engine, increasing click-through rates by 25% and average order value by 15%, and implemented A/B testing to optimize performance.
- Improved demand forecasting accuracy by 30% using machine learning, significantly enhancing inventory management and reducing stockouts.

- Reduced equipment downtime by 20% and maintenance costs by 15% through the implementation of machine learning for predictive maintenance.
- Enhanced product quality by 10% and reduced defects by 20% by utilizing machine learning for automatic defect detection in manufacturing processes.

2014/09 – 2015/10
Austin, Texas, United States

Machine Learning Engineer
The University of Texas at Austin

2013/05 – 2013/07

SDE Intern
Amazon

Education

2010 – 2015

Master's degree
The University of Texas at Dallas

Skills

Generative AI	Javascript
RAG	TensorFlow
PyTorch	Machine Learning
Computer Vision	Natural Language Processing (NLP)
Predictive Analytics	Advanced Analytics Tools
Model Optimization	Data Analytics
A/B Testing	Demand Forecasting
Predictive Maintenance	Optical System Design
Quality Control Systems	Research and Development
Technical Solution Evaluation	Compliance Management
Cost-Benefit Analysis	Calibration Automation
System Evaluation	

Languages

- English

Certificates

- | | |
|---|---|
| • Python for Data Science, AI & Development - IBM | • IBM Data Science Professional Certificate - IBM |
| • Visual Perception for Self Driving - University of Toronto3 | • TensorFlow Developer Certificate - TensorFlow Certificate Program |
| • AWS Certified Solutions Architect-Associate - Amazon Web Services (AWS) | |